

E-Commerce Sales Forecasting Using Machine Learning

Predicting monthly revenue with time-series regression

Samarth Singh

Aspiring Data Analyst

The Problem

Retail businesses need a read on next month's revenue to plan inventory, staffing, and budgets. Without it, decisions become reactive instead of proactive.

Before (Without forecasting)

- Reactive decisions based on guesswork
- Inventory mismatches — overstock or stockouts
- Budgets set without a data-backed baseline

After (With forecasting)

- Proactive planning using historical patterns
- Inventory aligned to seasonal demand
- Forecasts feed directly into budget planning

The Dataset

Superstore e-commerce transaction records, 2014–2017

9,994

Transactions

21 → 11

Columns after cleaning

4 years

2014 – 2017 span

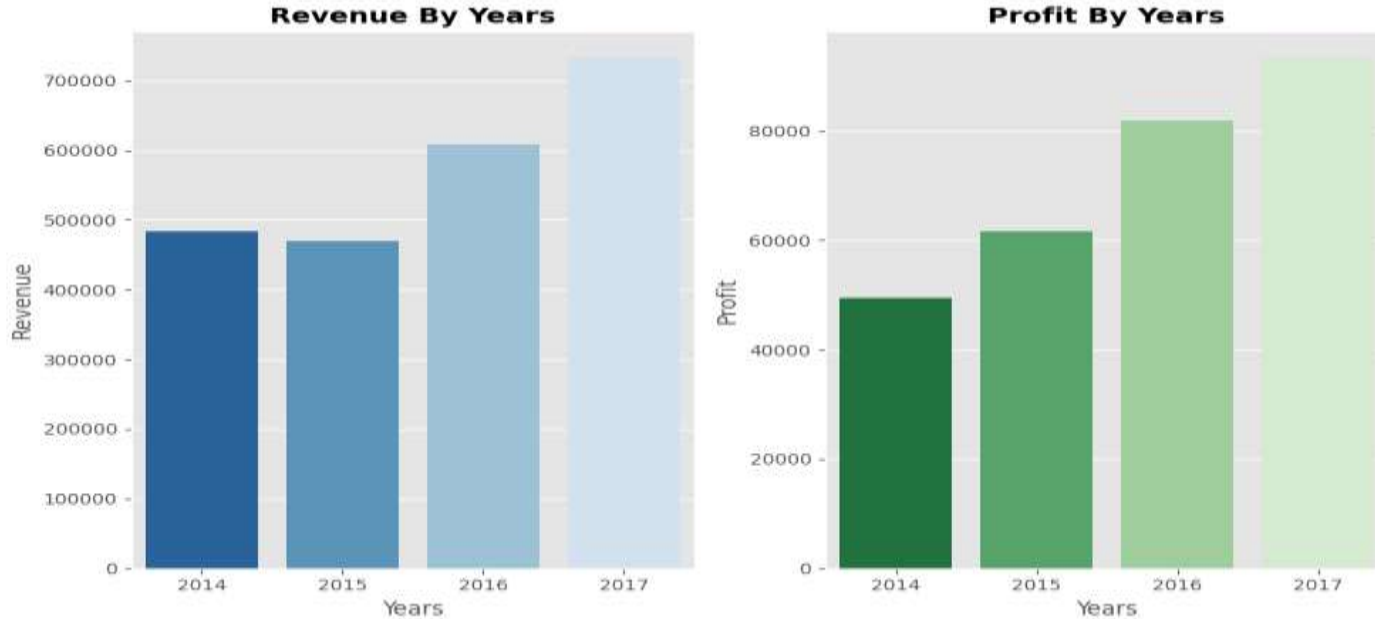
0

Duplicate & Null records

Customer Segments

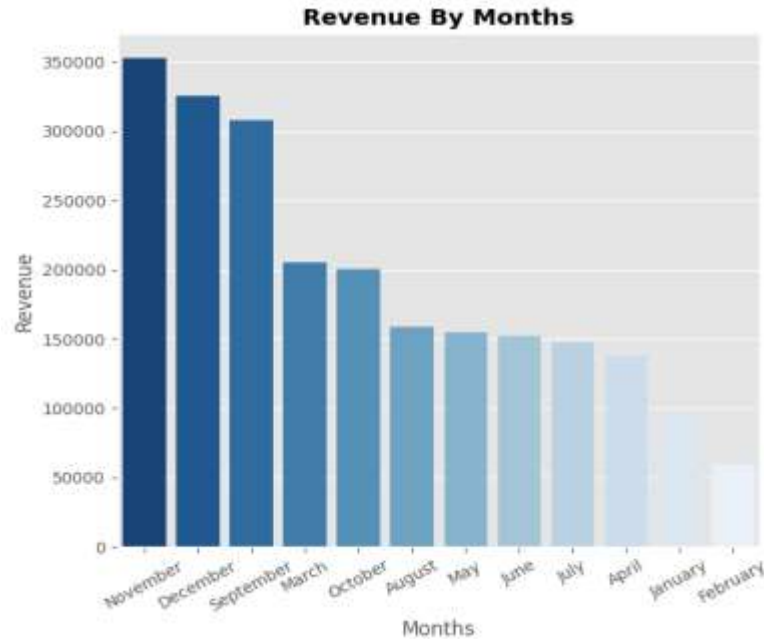
- Consumer — 5,191 orders (52%)
- Corporate — 3,020 orders (30%)
- Home Office — 1,783 orders (18%)
- Dropped non-predictive identifiers: Row ID, Customer ID, Postal Code, Ship Date, Ship Mode, and others

Revenue is growing year over year



Revenue and profit both climbed every year from 2014 to 2017, with the steepest jump between 2016 and 2017 — the underlying growth trend the model needs to pick up on.

Sales peak sharply at year-end

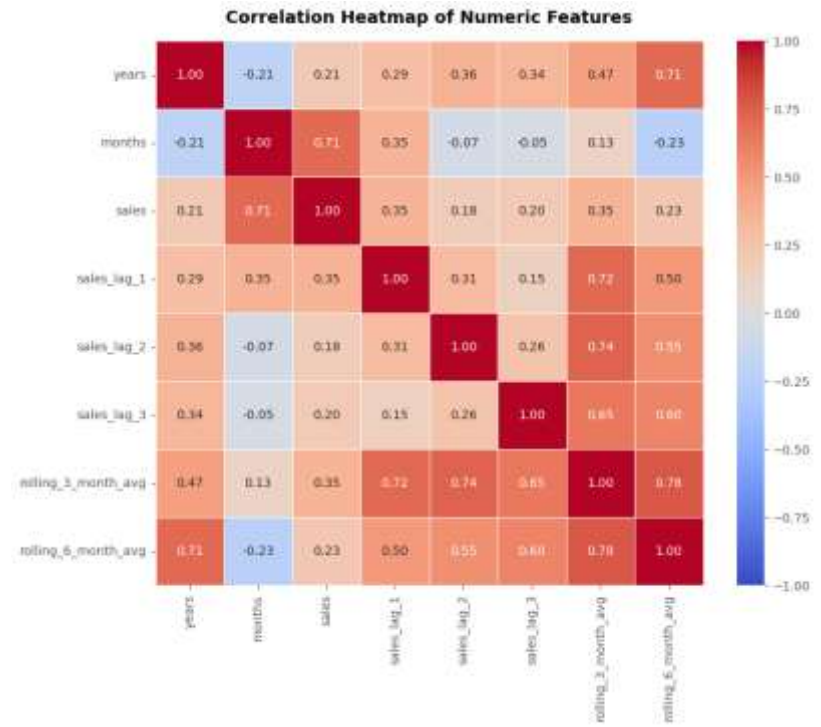


November, December, and September consistently outperform every other month — a clear, repeatable seasonal pattern the forecast needs to capture.

Giving the model memory



- Aggregated daily transactions into 48 monthly totals
- Added lag features: sales from 1, 2, 3 months ago
- Added 3-month and 6-month rolling averages
- Dropped early NaN rows — 42 usable months remain



Rolling averages and lag features correlate strongly with sales :- exactly what we want from engineered features.

Modeling approach

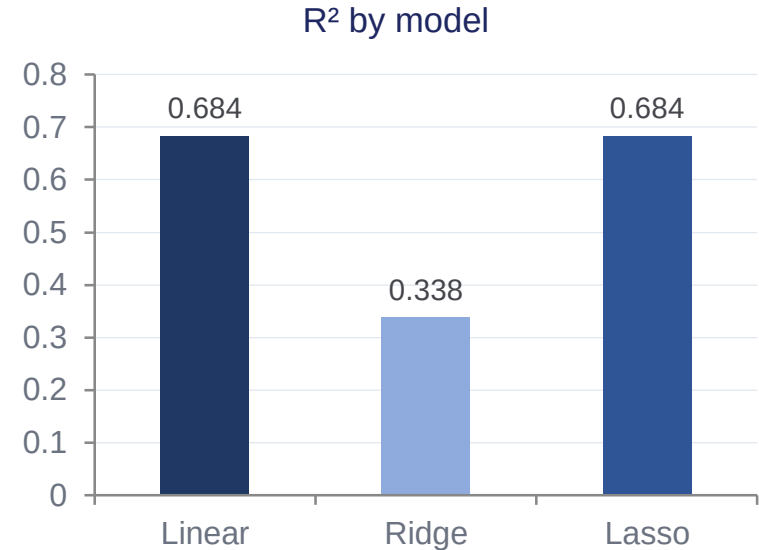
- Split chronologically, not randomly — order matters in time series
- Scaled lag and rolling features with StandardScaler
- One-hot encoded month to let the model learn seasonality
- Trained three models to compare: Linear, Ridge, Lasso



42 usable months, split 80 / 20 along the timeline

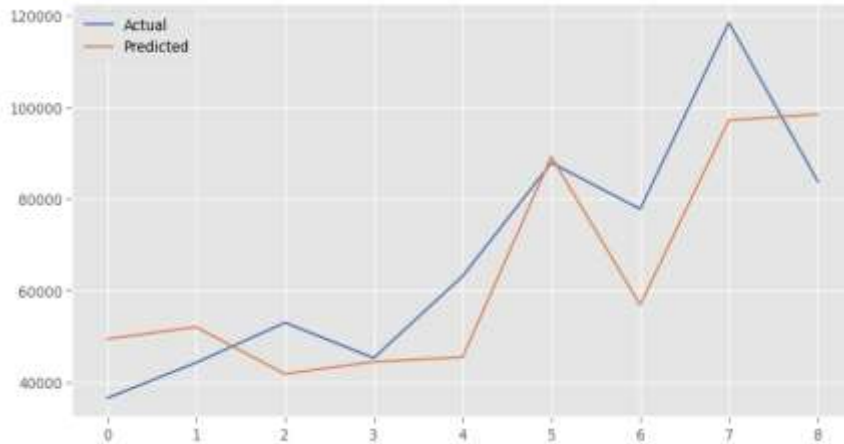
Linear-Regression wins

Model	MAE	RMSE	MAPE	R ²
Linear Regression	12,054.68	14,026.52	18.62%	0.684
Ridge	15,820.27	20,300.11	21.28%	0.338
Lasso	12,054.03	14,029.06	18.62%	0.684

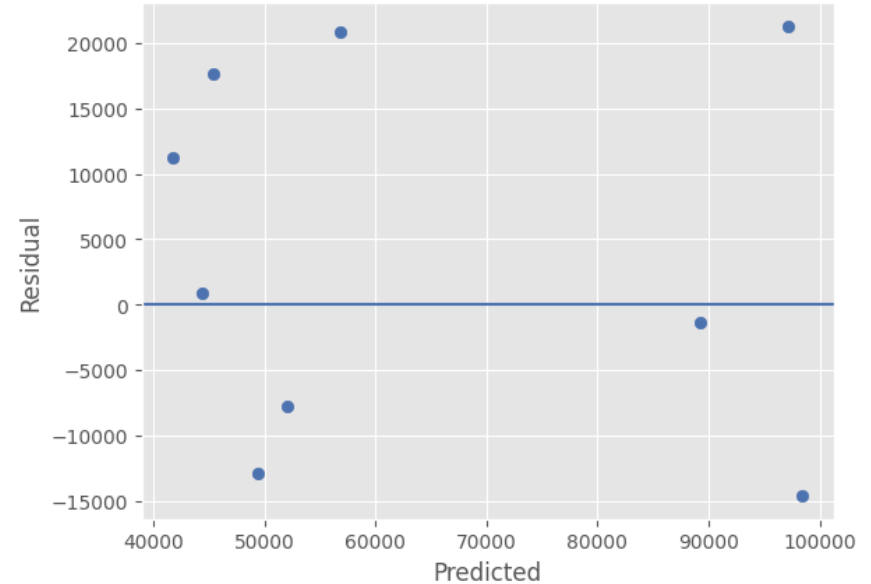


Linear and Lasso perform almost identically and best overall. Ridge underperformed — likely due to inconsistent feature scaling, a known and fixable limitation.

How well does it actually predict?



Predicted values track the overall trend closely across the test months.



Residuals center near zero with no strong pattern, though a few peak months see larger error.

What everybody should know

- Long-term upward sales trends across the 4 year period.
- Strong, repeatable monthly seasonality – certain months(Sept, Nov, Dec).
- Model gives a reasonable revenue estimate for budgeting and inventory planning.
- Recommendation: Increase inventory ahead of historically strong months
- Recommendation: Use monthly forecasting as a budgeting input, not a hard guarantee

What this means for the business



Steady growth

Revenue grew every year from 2014 to 2017.



Strong seasonality

Sept, Nov, and Dec consistently outperform.



Stock smarter

Increase inventory ahead of historically strong months.



Use as a planning input

Treat forecasts as budgeting guidance, not a guarantee.

Limitations & next steps

Limitations

- Test set is only 9 months — results carry real variance
- Model underestimates extreme monthly spikes

Next steps

- Add promotions, holidays, and marketing spend as features
- Compare against SARIMA or Prophet for seasonality

Thank you

[GitHub: github.com/OfficialSamarthSingh](https://github.com/OfficialSamarthSingh)

[LinkedIn: linkedin.com/in/samarth-singh-062371215](https://www.linkedin.com/in/samarth-singh-062371215)

[Portfolio: samarthsinghportfolio.netlify.app](https://samarthsinghportfolio.netlify.app)